NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Hand & Upper Limb

1. flexor and extensor tendons
2. nerve compression and injury
3. brachial plexus

flexor and extensor tendons

Flexor tendons

- **Zones (Verdan):**
  - I: distal to the FDS insertion (FDP only).
  - **II: "no man's land"**, from the A1 pulley to the FDS insertion (FDS and FDP in one sheath).
  - III: palm. IV: carpal tunnel. V: forearm.
  - Thumb: T1-T3.
- **Pulleys:** annular A1, A3 and A5 lie over joints. **A2 and A4 are the most important**: losing them causes bowstringing.
- **Repair principles:**
  - Repair strength rises with the **number of core strands** crossing the repair: use 4 or more strands plus an epitendinous suture.
  - Repair within about 2 weeks.
  - Start **early active motion** protocols.
  - Repairs are weakest at about **days 6-12**, the classic rupture window.
- **FDP avulsion (Jersey finger): ring finger** in about 75%. **Leddy-Packer:**
  - Type I: retracted to the palm, both vincula torn; repair within 7-10 days.
  - Type II: held at the PIP level by the long vinculum; repair possible up to about 6 weeks.
  - Type III: large bony fragment caught at A4; ORIF.
  - Type IV: bony fragment, with the tendon also avulsed from it.
- **Quadriga effect:** FDP tendons share a muscle belly, so advancing or tethering one FDP limits flexion of the neighbours.

- **Lumbrical-plus finger:** FDP divided distal to the lumbrical origin; trying to flex the finger extends the IP joints paradoxically.
- **Trigger finger:** A1 pulley; steroid injection first (works less well in diabetics); open or percutaneous A1 release.

Extensor tendons

- **Zones I-VIII:** odd zones lie over joints (I DIP, III PIP, V MCP, VII wrist). Thumb: T I-V.
- **Mallet finger (zone I):**
  - Splint the DIP in extension **continuously for 6-8 weeks**, then at night.
  - Surgery for **volar subluxation of the distal phalanx**; large articular fragments are debated.
  - An untreated mallet can lead to **swan-neck deformity**.
- **Boutonniere (zone III):**
  - Central slip rupture lets the lateral bands slip volar to the PIP axis.
  - The **Elson test** detects it early, before the deformity is fixed.
  - Splint the **PIP in extension for 6 weeks with the DIP free** to flex (keeps the lateral bands mobile).
- **Sagittal band rupture (boxer's knuckle):** most often the middle finger, radial band; the extensor tendon subluxes ulnarward.
- **EPL rupture after an undisplaced distal radius fracture:** attritional, at Lister's tubercle (watershed area); treat with **EIP-to-EPL transfer**.

Tenosynovitis around the wrist

- **De Quervain:** first dorsal compartment (**APL and EPB**). Finkelstein (or Eichhoff) test. An EPB in a separate subsheath

can explain a failed injection. Protect the **superficial radial nerve** at release.

- **Intersection syndrome:** where APL/EPB cross ECRL/ECRB (second compartment), about 4-6 cm proximal to Lister's tubercle; crepitus in rowers.
- **Dorsal compartments:** 1 APL/EPB; 2 ECRL/ECRB; 3 EPL; 4 EDC/EIP (with the PIN); 5 EDM; 6 ECU.

nerve compression and injury

Nerve injury

Seddon	Sunderland	Pathology	Recovery
Neurapraxia	I	Conduction block, axon intact	Complete, weeks
Axonotmesis	II-IV	Axon (and progressively endoneurium, perineurium) disrupted, epineurium intact	II-III good; IV needs surgery
Neurotmesis	V	Complete transection	Needs repair

- Mackinnon added a mixed **type VI**.
- **Wallerian degeneration** distal to the injury. Regeneration is about **1 mm/day (1 inch a month)**; an **advancing Tinel sign** tracks it.
- Motor end plates deteriorate after about **12-18 months**, so motor recovery needs reinnervation before then. This sets the timing of repair, grafts and transfers.
- **Repair:** tension-free epineural (or group fascicular) repair. Use a **sural nerve graft** for gaps; **conduits** only for short sensory gaps (under about 3 cm).

Median nerve

- **Carpal tunnel syndrome** (commonest compression neuropathy):
  - Night pain and paraesthesia in the radial 3.5 digits. Durkan compression, Phalen and Tinel tests.
  - **Thenar sensation is spared**: the palmar cutaneous branch runs superficial to the transverse carpal ligament. Sensory

loss over the thenar eminence points to a more **proximal** lesion.
    - Nerve conduction: prolonged **distal motor latency** (over about 4.2-4.5 ms) and slowed sensory conduction.
    - Treat with a neutral night splint and steroid injection, then **open or endoscopic release**. The recurrent motor branch is usually extraligamentous.
- **Pronator syndrome:** aching in the proximal forearm, sensory loss including the palm. Compression sites: ligament of Struthers, **lacertus fibrosus**, between the two heads of **pronator teres**, FDS arch. Nerve studies are often normal.
- **AIN syndrome: pure motor** (FPL, FDP to index and middle, pronator quadratus). The patient **cannot make an "OK" sign** (pinch becomes pulp-to-pulp).

Ulnar nerve

- **Cubital tunnel** (second commonest):
    - Compression sites: arcade of Struthers, medial intermuscular septum, medial epicondyle, **Osborne ligament** (commonest), between the FCU heads, deep flexor-pronator aponeurosis.
    - Signs: **Froment** (FPL substitutes for adductor pollicis), **Wartenberg sign** (abducted little finger), Jeanne sign, intrinsic wasting.
    - The **dorsal ulnar cutaneous branch is involved**, which distinguishes it from Guyon's canal.
    - **Ulnar paradox:** clawing is worse with a distal lesion (FDP to the ring and little fingers is spared).
    - **In situ decompression** gives results similar to anterior transposition. Transpose if the nerve subluxes.

- **Guyon's canal zones:** I mixed motor and sensory (proximal to the bifurcation); II motor only (deep branch; **hook of hamate** fracture, ganglion); III sensory only. A **ganglion** is the commonest cause. Hypothenar hammer syndrome: ulnar artery thrombosis.

Radial nerve

- **PIN syndrome: motor only**, finger and thumb drop. The wrist extends with **radial deviation**, because ECRL is spared and ECU is weak.
  - Compression sites: fibrous bands at the radiocapitellar joint, **leash of Henry**, ECRB edge, **arcade of Frohse** (commonest), distal supinator edge.
  - **Radial tunnel syndrome** is pain at the same sites without weakness.
- **Wartenberg syndrome:** superficial radial nerve between brachioradialis and ECRL, worse with pronation.
- **Radial palsy with a humeral shaft fracture** (classically the distal-third spiral **Holstein-Lewis** fracture): mostly neurapraxia. **Observe 3-4 months**; explore early for open fractures, and later if there is no recovery.
- **Saturday night palsy** (spiral groove): wrist drop with **triceps spared**.

brachial plexus

Anatomy

- Roots C5-T1 (prefixed from C4, postfixed to T2), trunks (upper C5-6, middle C7, lower C8-T1), divisions (behind the clavicle), cords (lateral, posterior, medial, named by position around the axillary artery), terminal branches.
- **Branches from the roots:** dorsal scapular (C5, rhomboids), **long thoracic (C5-7, serratus anterior)**, and the phrenic nerve contribution (C3-5).
- **Upper trunk branches:** suprascapular, nerve to subclavius.

Preganglionic (root avulsion) vs postganglionic injury

- **Signs of a preganglionic lesion:**
  - **Horner syndrome** (T1 sympathetic fibres).
  - **Winged scapula** (long thoracic) and rhomboid paralysis (dorsal scapular).
  - Raised hemidiaphragm (phrenic).
  - Severe burning **neuropathic pain**.
  - **No Tinel sign** in the neck.
  - **Pseudomeningocele** on CT myelography or MRI.
  - **Preserved SNAPs in an anaesthetic limb**: the dorsal root ganglion cell body is still attached to its peripheral axon. This is the classic electrodiagnostic sign.
- Preganglionic injuries **cannot be repaired**; function is restored with **nerve transfers**. Postganglionic ruptures can be grafted.

Patterns

- **Erb (C5-6):** "waiter's tip": adducted, internally rotated shoulder, extended elbow, pronated forearm.

- **Klumpke (C8-T1):** clawing and intrinsic paralysis, often with Horner syndrome.
- **Total plexus:** flail, anaesthetic arm.

Adult traction injuries (motorcycle accidents)

- **Timing:** sharp open injuries need immediate exploration and repair. Closed injuries: observe for about **3 months** for signs of recovery, but reconstruct **before about 6 months**, so reinnervation beats motor end-plate loss.
- **Priorities: elbow flexion** first, then **shoulder abduction and external rotation and stability**, then hand function.
- **Key nerve transfers:**

Goal	Transfer
Elbow flexion	**Oberlin:** an ulnar nerve fascicle (to FCU) to the biceps branch of the musculocutaneous nerve. **Double fascicular (Mackinnon):** add a median fascicle to the brachialis branch
Shoulder abduction / external rotation	**Spinal accessory to suprascapular** nerve
Deltoid	**Long head of triceps branch of the radial nerve to the axillary nerve**
Total plexus	Intercostals or contralateral C7 to musculocutaneous and other targets

- **Late presentation (over 12-18 months):** tendon transfers, **free functioning gracilis** muscle transfer, shoulder arthrodesis.

Obstetric brachial plexus palsy

- Risk factors: macrosomia, **shoulder dystocia**, assisted delivery. Most are Erb type and most recover.
- **Return of biceps function by 3 months** predicts good recovery (Gilbert-Tassin). If biceps has not recovered by **3-6 months**, explore with grafting or nerve transfers. Narakas grades severity.
- Sequelae: **internal rotation contracture** leading to posterior subluxation and **glenohumeral dysplasia**; assessed with the **Mallet score**.
- Treatment: release of subscapularis, and **latissimus dorsi and teres major transfer** to the rotator cuff for external rotation (L'Episcopo-type procedures). Humeral derotation osteotomy in older children.
- Differentials: clavicle or humerus fracture (pseudoparalysis), septic shoulder.